ClipSAT : Math for AI & Data Science

Unit 1.01: Vectors & Embeddings in AI

Name: _____

Q1. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [10, -15]^T .

[ ] A: |mathbf{w}|_1 = sqrt{325}, ; |mathbf
[ ] B: |mathbf{w}|_1 = -5, ; |mathbf{w}|_2
[ ] C: |mathbf{w}|_1 = 25, ; |mathbf{w}|_2
[ ] D: |mathbf{w}|_1 = 25, ; |mathbf{w}|_2

Q2. Two document embeddings mathbf{a} = [1, 2]^T and mathbf{b} = [2, 1]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{5} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q3. Two document embeddings mathbf{a} = [2, 4]^T and mathbf{b} = [4, 2]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{20} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q4. Two document embeddings mathbf{a} = [3, 6]^T and mathbf{b} = [6, 3]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{45} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q5. Two document embeddings mathbf{a} = [4, 8]^T and mathbf{b} = [8, 4]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{80} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q6. Two document embeddings mathbf{a} = [5, 10]^T and mathbf{b} = [10, 5]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{125} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q7. Two document embeddings mathbf{a} = [6, 12]^T and mathbf{b} = [12, 6]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{180} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[ ] C: 0.800
[ ] D: 0.500

Q8. The coordinate diagram shows the parabolic curve f_1x = x^2 - 6x + 20 in the domain of Vectors & Embeddings in AI. By completing the square f_1x = x - h^2 + k , find the sum k + f_15 .

[ ] A: 26
[ ] B: 29
[ ] C: 15
[ ] D: 24